

ВАРИАНТ 1 Буренков Григорий #321

(N1) $\frac{dz}{dt} = ?$ $z = x \operatorname{tg}(\sqrt{\cos 2x} + 2\sqrt{y})$ $\begin{cases} x = te^t \\ y = e^{-t} \end{cases}$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt} \quad \frac{dx}{dt} = e^t + te^t \quad \frac{dy}{dt} = -e^{-t}$$

$$\begin{aligned} \frac{\partial z}{\partial x} &= (x)' \cdot \operatorname{tg}(\sqrt{\cos 2x} + 2\sqrt{y}) + x(\operatorname{tg}(\sqrt{\cos 2x} + 2\sqrt{y}))' = \\ &= 1 \cdot \operatorname{tg}(\sqrt{\cos 2x} + 2\sqrt{y}) + \frac{1}{\cos^2(\sqrt{\cos 2x} + 2\sqrt{y})} \cdot x \cdot \frac{1}{2\sqrt{\cos 2x}} \cdot (-2\sin 2x) = \\ &= \operatorname{tg}(\sqrt{\cos 2x} + 2\sqrt{y}) - \frac{x \sin 2x}{\cos^2(\sqrt{\cos 2x} + 2\sqrt{y}) \cdot \sqrt{\cos 2x}} \end{aligned}$$

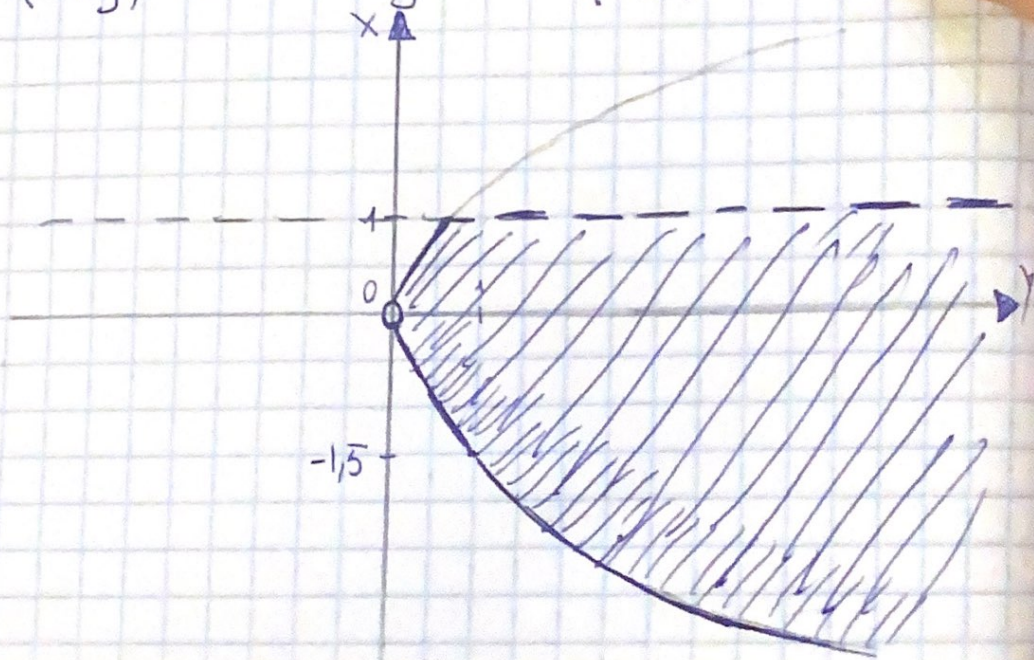
$$\frac{\partial z}{\partial y} = x \cdot \frac{1}{\cos^2(\sqrt{\cos 2x} + 2\sqrt{y})} \cdot 2 \cdot \frac{1}{2\sqrt{y}} = \frac{x}{\sqrt{y} \cos^2(\sqrt{\cos 2x} + 2\sqrt{y})}$$

Ответ: $\frac{dz}{dt} = (\operatorname{tg}(\sqrt{\cos 2x} + 2\sqrt{y})) \cdot (e^t + te^t) + \left(\frac{x}{\sqrt{y} \cos^2(\sqrt{\cos 2x} + 2\sqrt{y})} \right) \cdot e^{-t}$

(N2) $z = f(x, y) \quad z = \sqrt{4x - y^2} + \ln\left(\frac{1-y}{x}\right)$

ООП:

$$\begin{cases} 4x \geq y^2 \\ x \neq 0 \\ y < 1 \end{cases}$$



(N3) $z = x^2 + xy + y^3 - 2x - y$

$$\frac{\partial z}{\partial x} = 2x + y - 2$$

$$\frac{\partial z}{\partial y} = x + 3y^2 - 1$$

$$\begin{cases} 2x + y - 2 = 0 \\ x + 3y^2 - 1 = 0 \end{cases} \Rightarrow \begin{cases} y = 0; y = \frac{1}{6} \\ x = 1; x = \frac{11}{12} \end{cases}$$

поискать x в верх.

продолжение см. 3мсв $\Rightarrow (x_1; y_1) = (1, 0)$
 $(x_2; y_2) = \left(\frac{11}{12}; \frac{1}{6}\right)$

(N4) $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$ $z = \sin^2 \ln(x-y)$

$$\frac{\partial z}{\partial x} = -\frac{f'_x}{f'_z} \cdot \frac{\partial z}{\partial y} = -\frac{f'_y}{f'_z}$$

$$\frac{\partial z}{\partial x} = \frac{\sin 2 \ln(x-y)}{x-y}$$

$$\frac{\partial z}{\partial y} = -\frac{\sin 2 \ln(x-y)}{x-y}$$

$$f(x, y, z) = z - \sin^2(\ln(x-y))$$

$$f'_x = -\frac{\sin(2 \ln(x-y))}{x-y}$$

$$f'_y = \frac{\sin(2 \ln(x-y))}{x-y}$$

$$f'_z = 1$$

$$\frac{\sin 2 \ln(x-y)}{x-y} - \frac{\sin 2 \ln(x-y)}{x-y} = 0$$

г.т.г

N5 $z = x^3 - 3xy^2 + 18y$ $M(1; 1)$

$$\begin{cases} f'_x = 3x^2 - 3y^2 = 6 \\ f'_y = 6xy + 18 = 24 \end{cases} \Rightarrow \vec{\text{grad}} z = 6\vec{i} + 24\vec{j}$$

скорость возрастания функции $|\vec{\text{grad}} z| = \sqrt{36 + 24^2} = 2\sqrt{153}$

направление наибольшей скорости $\parallel \vec{\text{grad}} z \parallel = \frac{13}{\sqrt{153}} \vec{i} + \frac{12}{\sqrt{153}} \vec{j}$

№3

Продолжение, мы нашли стационарные точки, теперь

$$Z''_{xx} = 2$$

$$Z''_{xy} = 1$$

$$Z''_{yy} = 6y$$

$$\Delta = AC - B^2 = 2 \cdot 6y - 1 = 12y - 1 = 2 - 1 = 1$$

подставим $\left(\frac{11}{12}, \frac{1}{6}\right)$

$$Z = \left(\frac{11}{12}\right)^2 + \frac{11}{12} \cdot \frac{1}{6} + \left(\frac{1}{6}\right)^3 - 2 \cdot \frac{11}{12} - \frac{1}{6} = -\frac{433}{432}$$